3.1.1 Show natural projection $f: X(\Gamma_1) \to X(\Gamma_2)$ $\quad \Gamma_1 \subset \Gamma_2$

$\Gamma_1\tau \mapsto \Gamma_2\tau$

has degree claimed in the text:

$$\deg(f) = [\{\pm I\}\Gamma_2 : \{\pm I\}\Gamma_1] = \begin{cases} [\Gamma_2 : \Gamma_1]/2 & -I \in \Gamma_2,\ -I \notin \Gamma_1 \\ [\Gamma_2 : \Gamma_1] & \text{otherwise} \end{cases}$$

Pf: let $\{\pm I\}\Gamma_2 = \bigcup_j \{\pm I\}\Gamma_1 \gamma_j$. So we want $\deg(f) = j$.

~~Then~~ From definition $f^{-1}(\Gamma_2\tau) = \{\Gamma_1\gamma_j\tau : j\}$.

For $y = \Gamma_2\tau$ not an elliptic point

that is $|\{\gamma \cancel{\tau = \tau} : \cancel{\gamma \in \Gamma_2}\ \gamma\tau = \tau\}| = 1$,

$|\{\Gamma_1\gamma_j\tau : j\}| = j$.

Thus $\deg(f) = |f^{-1}(y)| = j$. $\quad$ except for finite elliptic points y.

not elliptic

3.1.2. Show if $\Gamma_1 \trianglelefteq \Gamma_2$, then Given y, $x \in X(\Gamma_1)$, $y \in X(\Gamma_2)$,

$\forall x \in f^{-1}(y)$ $\quad e_x$ is the same.

Pf: Recall ramification degree

$$e_{\pi_1(\tau)} = h_2/h_1 = \begin{cases} h_2 & \tau \text{ elliptic point for } \Gamma_2 \text{ but not } \Gamma_1 \\ 1 & \text{otherwise} \end{cases} = [\{\pm I\}\Gamma_{2,\tau} : \{\pm I\}\Gamma_{1,\tau}]$$

We only need $[\{\pm I\}\Gamma_{2,\gamma(\tau)} : \{\pm I\}\Gamma_{1,\gamma(\tau)}] = [\{\pm I\}\Gamma_{2,\tau} : \{\pm I\}\Gamma_{1,\tau}]$

$\forall \gamma \in \Gamma_2$..

By definition, $\{\pm I\}\Gamma_{2,\gamma(\tau)} = \{\gamma' \in \Gamma_2 : \gamma'(\gamma(\tau)) = \gamma(\tau)\}$.

$= \{\gamma^{-1}\gamma'\gamma \in \Gamma_2 : \gamma^{-1}\gamma'\gamma(\tau) = \tau\}$

$= \gamma^{-1}\{\pm I\}\Gamma_{2,\tau}\gamma$

Thus LHS $= [\gamma^{-1}\{\pm I\}\Gamma_{2,\tau}\gamma : \gamma^{-1}\{\pm I\}\Gamma_{1,\tau}\gamma] = [\{\pm I\}\Gamma_{2,\tau} : \{\pm I\}\Gamma_{1,\tau}]$

3.1.3 (a) Fill in details in the computations preceding Theorem 3.1.1.

Sol: Now we have $f : X(\Gamma) \to X(1)$.

$y_2 = SL_2(\mathbb{Z})\, i$

$y_3 = SL_2(\mathbb{Z})\, \rho_3$

$y_\infty = SL_2(\mathbb{Z})\, \infty$

$$\varepsilon_2 = |\{\tau \in f^{-1}(y_2) : \Gamma\tau = \tau\}|$$

$$\varepsilon_3 = |\{\tau \in f^{-1}(y_3) : \Gamma\tau = \tau\}|$$

$$\varepsilon_\infty = |\{\tau \in f^{-1}(y_\infty)\}|.$$

Then $d = \sum_{x \in f^{-1}(y)} e_x =$